

# Abiodun M. Musbaudeen

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## Education

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| <b>Ph.D. in Economics</b> , Binghamton University (SUNY)                | Expected May 2027 |
| Dissertation: <i>Essays on Digitalization, Productivity, and Markup</i> |                   |
| Committee: Subal C. Kumbhakar (chair), Ivan Korolev, Yingjun Su         |                   |
| <b>M.Sc. in Economics</b> , University of Lagos, Nigeria                | 2017              |
| <b>B.Sc. in Economics</b> , University of Lagos, Nigeria                | 2014              |

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## Research Interests

Applied Microeconomics; Production Econometrics; Industrial Organization.

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## Honors and Awards

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| <b>Best Third-Year Paper Award (co-recipient)</b> , Department of Economics, Binghamton University | 2025      |
| Graduate Teaching and Research Assistantship, Binghamton University                                | 2022–2027 |

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## Publications

### Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects

Ivan Korolev, Abiodun Musbaudeen, and Nancy Dhameja. 2026.  
*Economics Letters* 268: 113205. doi:10.1016/j.econlet.2026.113205

### The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking

Abiodun M. Musbaudeen and Subal C. Kumbhakar. 2026.  
*Economic Modelling*, 107859. Journal article; replication code.  
A control-function input distance model for 17 banks over 2000–2023 separates productivity from technical change. Productivity declined by 3.5% annually on average despite cost-reducing technical change and estimated returns to scale of 1.46.

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## Job Market Paper

### Digitalization, Factor-Biased Productivity, and Market Power

With Subal C. Kumbhakar.  
Develops a method to sequentially estimate labor-augmenting productivity, Hicks-neutral productivity, and markups, allowing for persistence and endogenous digitalization. Extends proxy-variable identification to firms with market power without imposing a particular model of competition. Applies the method to Spanish manufacturing firms (ESEE), 2000–2019, using a multidimensional digitalization index. Digitalization is associated with gains in both productivity components, with larger labor-augmenting responses for small firms and smaller markup responses in highly digitalized industries.

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## Working Papers

### Mediation Analysis in Production Functions with Unobserved Productivity

With Subal C. Kumbhakar.  
Develops a structural framework for direct and input-mediated effects when unobserved productivity confounds treatment, input choice, and output. Combines simulations with an application to Spanish manufacturing firms using the Encuesta Sobre Estrategias Empresariales (ESEE), 2000–2019 (estimation sample: 2001–2019).

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## Work in Progress

### Regularization in Translog Production Models

Solo-authored.  
Evaluates regularization methods by the accuracy and interpretability of production elasticities.

### Artificial Intelligence and Spatial Factor-Biased Productivity

Solo-authored.  
Studies how AI adoption changes factor productivity and how effects spread across firms and regions.

## Teaching Experience

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*Department of Economics, Binghamton University (SUNY)*

### Instructor

#### Statistical Methods

Summer 2025

I designed the syllabus, lectures, problem sets, data exercises, and assessments around the course's learning objectives, emphasizing statistical reasoning and the interpretation of results.

### Teaching Assistant

**Introduction to Econometrics**, Prof. Subal C. Kumbhakar

Fall 2026; Fall 2025

Lead econometrics TA sessions and R instruction; hold office hours.

**Advanced Data Analytics**, Prof. Ivan Korolev

Spring 2026

**Principles of Macroeconomics**

Spring 2025

**Corporate Economics**, Prof. Charles Sebhara

Fall 2024

**Principles of Microeconomics**, Prof. Amarendra Sharma

Spring 2024

**Introduction to Microeconomics**, Prof. Wenya Wang

Fall 2023

**Principles of Macroeconomics**, Prof. Kenny Christianson

Spring 2023

**Intermediate Microeconomics**, Prof. Amarendra Sharma

Fall 2022

### Presentations

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Southern Economic Association Annual Meeting (*scheduled*)

November 2026

North American Productivity Workshop (NAPW)

2025

### Professional Experience

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**Data Engineer and Business Intelligence Analyst**, Mindshare (WPP Media), New York

2021–2022

- Built 43 ETL pipelines consolidating international campaign and spending data; unified reporting across social-media and Adobe Analytics sources and automated Tableau and Power BI reports for global decision-makers.

**Head of Data Science and Engineering**, Konga eCommerce Group, Nigeria

2021

- Led a five-person data science and engineering team; directed machine-learning customer segmentation and translated results into targeting and engagement recommendations for senior management.

**Senior Data Scientist and Data Engineer**, First Bank of Nigeria Ltd., Nigeria

2018–2021

- Built more than 50 data pipelines and applied machine learning to segment approximately 23 million customer accounts; developed dormancy and channel-choice models to inform retention and service decisions.

**Data Research Analyst**, West Africa Vocational Education, Nigeria

2018

- Analyzed trainee retention and employment outcomes to inform admissions decisions; designed questionnaires and compared outcomes for program and non-program candidates.

### Computing and Methods

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**Programming:** R, Python, Stata, SQL,  $\text{\LaTeX}$ , Git.

**Data and visualization:** Snowflake, Oracle, Azure Data Factory, Azure Synapse Analytics, Tableau, Power BI.

**Methods:** GMM, instrumental variables, control functions, production and distance functions, stochastic frontier analysis, panel data, causal mediation, causal forests, difference-in-differences, synthetic control, regression discontinuity, bootstrap inference, simulation, and regularization.

### References

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**Subal C. Kumbhakar**

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**Ivan Korolev**

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**Yingjun Su**

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